

Supporting Information

Beyond Aggregate Reliability in ADMET Prediction: Protecting Weakest-Class Coverage under Molecular Distribution Shift with a Transport-Audited Multi-View Envelope

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S1 Scope and frozen evidence boundary

This Supporting Information accompanies the two-stage ADMET reliability study reported in the main manuscript. Stage I is the frozen four-endpoint diagnostic audit. Development splits and seed 99 were excluded from its scientific conclusions. Random and scaffold analyses used confirmatory seeds 101–110, and similarity-cluster analyses used seeds 101–105. Model families were not treated as independent inferential replicates. Method contrasts were paired within the same endpoint, split, seed, model, and classification imbalance regime.

Stage II evaluates TAME under the prospectively separated RACER-C4 protocol. The public Tox21 leaderboard batch and seeds 101–105 were used for architecture development only. The independent EPA batch used fresh seeds 211–215. The executable lock, method, protected label, primary endpoints, promotion checks, efficiency guardrail, final-label hash, and negative-result policy were fixed before final-label access. Final predictions and the label-blind transport audit were hashed into the promotion record before the locked label file was acquired or parsed.

The Stage I tables below are generated directly from the versioned, integrity-checked manuscript result package. The generation step performs no model fitting, split construction, conformal threshold estimation, method selection, or additional hypothesis testing. Descriptive cross-model averages are used for compact endpoint- and split-level reporting. Where inferential stability is discussed, paired-effect tables report the number of frozen model-specific cross-seed 95% confidence intervals that lie entirely above or below zero in the stated direction. The Stage II tables are transcribed from the frozen publication report, endpoint summary, transport-audit summary, and exclusion record.

S2 Reproducibility and machine-readable files

From the repository root, the Supporting Information tables and PDF are generated with:

```
conda activate aidd_paper
python paper2_admet_benchmark/scripts/36_build_clean_supporting_information.py
cd paper2_latex
latexmk -C supplementary.tex
latexmk -pdf -interaction=nonstopmode -file-line-error -halt-on-error supplementary.tex
```

The clean builder reads only the frozen CSV files under `paper2_admet_benchmark/results/manuscript_ as sets/`. It does not require local row-level prediction files, trained model binaries, or intermediate aggregation tables. Its post-processing is limited to LaTeX safety, readable display labels, landscape-page placement, and print formatting; it does not alter any numerical result. The machine-readable CSVs remain the authoritative source for full decimal precision, complete SHA-256 hashes, and secondary metrics. High-domain ClinTox subgroup results under scaffold and similarity-cluster splitting should be treated as descriptive because positive-class support can be sparse. No sparse or adverse result was removed after inspection.

S3 Frozen TAME method and RACER-C4 evaluation specification

Table S1: **Prospectively separated data roles and evaluation units.**

Cohort	Seeds	Locked role
Tox21 10K training	Per run	60% fit, 20% logit router, 20% internal class-conditional conformal calibration
Public leaderboard (296 structures)	101–105	Architecture development and promotion gate only
Final EPA file (647 structures)	211–215	One label-firewalled independent evaluation

Table S2: **Complete transport certificate applied separately to each candidate view.** A view was active only when every row passed.

Diagnostic	Frozen acceptance threshold
Source and target sample size	Both ≥ 100
Internal conformal support	Each true class ≥ 20
Total and per-class effective sample size	Each $\geq 25\%$ of raw count
Weights at lower or upper clipping bound	Each fraction $\leq 30\%$
Five-fold cross-fitted domain ROC–AUC	≤ 0.99
Weighted/unweighted standardized mean-gap ratio	≤ 1.05

The approved views were nine physicochemical descriptors and five component/router logits. A 2048-bit ECFP view was audited as a mandatory diagnostic but was ineligible for the envelope after failing public development. Working odds were prior-corrected by the source-to-target sample-size ratio and clipped to $[0.05, 20]$. If both approved views passed, TAME could add label 0 only when both corresponding weighted Mondrian sets contained label 0. It could never remove an ordinary baseline label. If either view failed, the transport augmentation was disabled. An ordinary empty set was converted to $\{0, 1\}$ in either branch.

S4 Public development gate

Table S3: **Frozen TAME promotion checks on six primary endpoints by five development seeds.**

Quantity	Observed	Gate result
Completed primary endpoint–seed cells	30/30	Pass
Mean class-0 coverage change	+5.4394 pp	Pass
Mean class-1 coverage change	0.0000 pp	Pass
Mean MacroCSY change	−4.9169 pp	Pass (≥ -5 pp)
Baseline-inclusion violations	0	Pass
TAME empty sets	0	Pass
Newly created singletons	0	Pass
Both-view activity	24/30	Pass ($\geq 80\%$)

The score view passed 30/30 development cells and the physicochemical view 24/30. The exact ordinary fallback handled the other six. The ECFP diagnostic passed 0/30 because of effective-sample-size and/or clipping failures. These values are architecture-development evidence and were not pooled with independent results.

S5 Independent primary endpoint results

Table S4: **Endpoint means for TAME minus ordinary routed Mondrian prediction on the final EPA cohort.** Each row averages five fresh seeds. Values are percentage points.

Endpoint	Minimum-class coverage	MacroCSY	Class-0 coverage
NR-AhR	+2.1644	−1.8571	+2.5049
NR-ER	+1.2362	−0.2128	+1.2362
SR-ARE	+2.1461	−2.6374	+2.3982
SR-ATAD5	0.0000	−1.4286	+1.4363
SR-MMP	+1.1489	−2.2222	+1.1489
SR-p53	+1.4936	−1.2821	+1.4936
Primary endpoint–seed-equal mean	+1.3649	−1.6067	—

The primary 95% interval was +0.5827 to +2.0051 percentage points. It used 2000 hierarchical-bootstrap draws with seed 44021, resampling endpoints first and compounds within endpoint and true class second while retaining all five seeds. The point estimate was positive and the MacroCSY point estimate was inside the frozen −5-point efficiency guardrail. On the class-balanced MacroCSY scale, the guardrail permits at most five fewer correct singleton outputs per 100 cases. It was fixed after public development as an architecture-selection tolerance and before final-label access; it was not calibrated to clinical or regulatory utility. The executable lock retains the historical field name `macro_csy_noninferiority_margin` and result code `coverage_gain_with_efficiency_noninferiority`. Because no MacroCSY confidence-interval criterion was prespecified, the manuscript does not claim formal non-inferiority. The lock also explicitly prohibited an arbitrary-shift coverage guarantee or automatic scientific-superiority claim.

S6 Final transport audit, invariants, and exclusions

Table S5: **Label-blind final transport activity across all 12 endpoints and five prospective seeds.**

View	Active cells	Frozen outcome
Physicochemical descriptors	60/60	Approved envelope view
Component and routed logits	60/60	Approved envelope view
ECFP bits	0/60	Diagnostic only; certificate failed

All 60 final endpoint–seed cells completed. Across development and final evaluation, TAME had zero baseline-inclusion violations, zero newly created singletons, and zero emitted empty sets. Two final external structures failed the frozen RDKit standardizer; their public sample identities were retained with full-set exclusions. Standardized overlaps with the 10K training cohort were excluded separately by endpoint, yielding 308 recorded endpoint–structure overlap pairs.

A post-evaluation code audit found that the original percentile bootstrap iterated unique drawn endpoints through an unordered Python set. A hash-verified inference-only repair preserved first-occurrence order. It changed only the percentile interval, from the original sealed [+0.5827, +1.9696] points to the deterministic publication interval [+0.5827, +2.0051] points after rounding. Predictions, labels, metric tables, the +1.3649-point estimate, the −1.6067-point MacroCSY change, and the predeclared interpretation were unchanged. The original and repaired reports are both retained.

S7 Stage I generated audit tables

S8 Predictive performance and calibration

Table S6: Descriptive classification performance means across the frozen model-regime and seed cells. Model families are not inferential replicates.

Endpoint	Split	ROC-AUC	PR-AUC	Balanced accuracy
bbbp	cluster	0.864	0.941	0.748
bbbp	random	0.886	0.949	0.789
bbbp	scaffold	0.851	0.940	0.736
clintox	cluster	0.675	0.274	0.552
clintox	random	0.703	0.283	0.575
clintox	scaffold	0.679	0.243	0.537

Table S7: Descriptive classification calibration means across the frozen model-regime and seed cells.

Endpoint	Split	Brier	NLL	Probability ECE	Confidence ECE
bbbp	cluster	0.119	0.515	0.088	0.076
bbbp	random	0.102	0.439	0.075	0.063
bbbp	scaffold	0.120	0.525	0.096	0.082
clintox	cluster	0.065	0.390	0.053	0.049
clintox	random	0.060	0.323	0.051	0.043
clintox	scaffold	0.068	0.379	0.057	0.052

Table S8: Descriptive regression performance means across the frozen model and seed cells.

Endpoint	Split	RMSE	MAE	R^2
esol	cluster	1.508	1.158	0.451
esol	random	1.441	1.091	0.488
esol	scaffold	2.035	1.613	0.089
lipophilicity	cluster	1.386	1.049	-0.617
lipophilicity	random	1.192	0.879	-0.175
lipophilicity	scaffold	1.321	1.005	-0.503

S9 Classification conformal prediction

Table S9: Classification conformal summaries at $\alpha = 0.10$. Values are descriptive averages across frozen model-regime and seed cells.

Endpoint	Split	Method	Coverage	Positive cov.	Negative cov.	Class gap	Mean set size	Empty rate	Ambiguous rate
bbbp	cluster	Density-ratio weighted LAC	0.905	0.973	0.689	0.284	1.152	0.001	0.153
bbbp	cluster	Marginal LAC	0.907	0.973	0.694	0.279	1.144	0.000	0.145
bbbp	cluster	Mondrian LAC	0.904	0.898	0.920	0.038	1.377	0.000	0.377
bbbp	random	Density-ratio weighted LAC	0.907	0.972	0.699	0.272	1.089	0.001	0.090
bbbp	random	Marginal LAC	0.904	0.971	0.689	0.281	1.076	0.000	0.076
bbbp	random	Mondrian LAC	0.905	0.907	0.898	0.036	1.289	0.000	0.289
bbbp	scaffold	Density-ratio weighted LAC	0.907	0.977	0.674	0.303	1.155	0.001	0.156
bbbp	scaffold	Marginal LAC	0.907	0.978	0.664	0.314	1.142	0.001	0.143
bbbp	scaffold	Mondrian LAC	0.901	0.903	0.886	0.064	1.350	0.000	0.350
clintox	cluster	Density-ratio weighted LAC	0.912	0.158	0.972	0.813	1.001	0.029	0.030
clintox	cluster	Marginal LAC	0.897	0.069	0.963	0.894	0.958	0.042	0.000
clintox	cluster	Mondrian LAC	0.902	0.926	0.900	0.066	1.709	0.000	0.709
clintox	random	Density-ratio weighted LAC	0.901	0.137	0.955	0.818	0.966	0.046	0.012
clintox	random	Marginal LAC	0.898	0.099	0.954	0.854	0.950	0.050	0.000
clintox	random	Mondrian LAC	0.899	0.940	0.896	0.085	1.719	0.000	0.719
clintox	scaffold	Density-ratio weighted LAC	0.890	0.073	0.954	0.881	0.960	0.049	0.009
clintox	scaffold	Marginal LAC	0.902	0.068	0.968	0.899	0.971	0.035	0.006
clintox	scaffold	Mondrian LAC	0.898	0.940	0.895	0.074	1.717	0.000	0.717

Table S10: Paired classification conformal effects. Deltas are treatment minus marginal conformal prediction on matched cells. CI-count columns report how many of the eight frozen model–regime cross-seed 95% intervals exclude zero in the stated direction.

Endpoint	Split	Comparison	Δ positive cov.	Model CIs > 0	Δ class gap	Model CIs < 0	Δ mean set size	Δ ambiguity
bbbp	cluster	Mondrian - marginal	-0.075	0	-0.241	8	0.232	0.232
bbbp	cluster	Shift-weighted - marginal	0.000	0	0.005	0	0.008	0.008
bbbp	random	Mondrian - marginal	-0.064	0	-0.245	8	0.212	0.212
bbbp	random	Shift-weighted - marginal	0.001	0	-0.009	0	0.013	0.014
bbbp	scaffold	Mondrian - marginal	-0.074	0	-0.250	8	0.208	0.207
bbbp	scaffold	Shift-weighted - marginal	-0.001	0	-0.011	0	0.013	0.014
clintox	cluster	Mondrian - marginal	0.857	8	-0.828	8	0.750	0.709
clintox	cluster	Shift-weighted - marginal	0.090	0	-0.081	0	0.043	0.030
clintox	random	Mondrian - marginal	0.841	8	-0.770	8	0.769	0.719
clintox	random	Shift-weighted - marginal	0.037	0	-0.036	1	0.016	0.012
clintox	scaffold	Mondrian - marginal	0.872	8	-0.825	8	0.746	0.711
clintox	scaffold	Shift-weighted - marginal	0.005	0	-0.018	1	-0.011	0.002

S10 Regression conformal prediction

Table S11: Regression conformal summaries at $\alpha = 0.10$. Width-error association is undefined for constant-width marginal intervals.

Endpoint	Split	Method	Coverage	Absolute gap	Mean width	Width CV	Width-error ρ
esol	cluster	Density-ratio weighted residual	0.907	0.029	5.063	0.044	0.053
esol	cluster	Marginal absolute residual	0.911	0.019	5.094	0.000	–
esol	cluster	Adaptive normalized	0.901	0.020	4.915	0.008	0.049
esol	random	Density-ratio weighted residual	0.893	0.026	4.648	0.024	-0.005
esol	random	Marginal absolute residual	0.901	0.021	4.817	0.000	–
esol	random	Adaptive normalized	0.900	0.023	4.776	0.007	0.053
esol	scaffold	Density-ratio weighted residual	0.897	0.040	6.916	0.052	0.059
esol	scaffold	Marginal absolute residual	0.891	0.065	6.892	0.000	–
esol	scaffold	Adaptive normalized	0.893	0.058	6.821	0.013	0.199
lipophilicity	cluster	Density-ratio weighted residual	0.903	0.017	4.724	0.009	-0.069
lipophilicity	cluster	Marginal absolute residual	0.892	0.018	4.491	0.000	–
lipophilicity	cluster	Adaptive normalized	0.891	0.020	4.478	0.032	0.102
lipophilicity	random	Density-ratio weighted residual	0.898	0.012	3.939	0.006	0.014
lipophilicity	random	Marginal absolute residual	0.899	0.007	3.922	0.000	–
lipophilicity	random	Adaptive normalized	0.899	0.011	3.901	0.032	0.152
lipophilicity	scaffold	Density-ratio weighted residual	0.906	0.020	4.535	0.006	-0.019
lipophilicity	scaffold	Marginal absolute residual	0.900	0.024	4.361	0.000	–
lipophilicity	scaffold	Adaptive normalized	0.900	0.025	4.342	0.033	0.115

Table S12: Paired regression conformal effects. Deltas are treatment minus marginal intervals on matched endpoint, split, seed, and model cells; CI counts are out of four frozen regressors.

Endpoint	Split	Comparison	Δ coverage	Δ absolute gap	Model CIs gap < 0	Δ mean width	Model CIs width > 0
esol	cluster	Adaptive - marginal	-0.010	0.000	0	-0.178	0
esol	cluster	Shift-weighted - marginal	-0.004	0.010	0	-0.031	0
esol	random	Adaptive - marginal	-0.001	0.002	0	-0.041	0
esol	random	Shift-weighted - marginal	-0.008	0.005	0	-0.169	0
esol	scaffold	Adaptive - marginal	0.001	-0.007	0	-0.071	0
esol	scaffold	Shift-weighted - marginal	0.006	-0.026	1	0.024	0
lipophilicity	cluster	Adaptive - marginal	-0.001	0.002	0	-0.013	0
lipophilicity	cluster	Shift-weighted - marginal	0.011	-0.001	0	0.234	1
lipophilicity	random	Adaptive - marginal	0.000	0.004	0	-0.021	0
lipophilicity	random	Shift-weighted - marginal	-0.001	0.005	0	0.017	0
lipophilicity	scaffold	Adaptive - marginal	0.000	0.001	0	-0.019	0
lipophilicity	scaffold	Shift-weighted - marginal	0.006	-0.004	0	0.174	1

S11 Applicability-domain sensitivity analyses

Table S13: Continuous applicability-domain diagnostics. Blank regression risk-OD AUC cells reflect the prespecified output definition rather than missing runs.

Endpoint	Split	Risk-similarity ρ	Miscoverage-similarity ρ	Risk OOD AUC	Miscoverage OOD AUC	Model CIs risk < 0	Model CIs miscov. < 0
bbbp	cluster	-0.131	-0.120	0.607	0.617	8	8
bbbp	random	-0.141	-0.127	0.623	0.624	8	8
bbbp	scaffold	-0.159	-0.125	0.630	0.625	8	8
clintox	cluster	-0.038	-0.056	0.543	0.551	1	3
clintox	random	-0.021	-0.039	0.525	0.537	1	4
clintox	scaffold	-0.053	-0.064	0.556	0.562	3	7
esol	cluster	-0.149	-0.094	–	0.601	1	1
esol	random	-0.164	-0.114	–	0.611	4	4
esol	scaffold	0.042	0.082	–	0.427	0	0
lipophilicity	cluster	-0.222	-0.188	–	0.677	4	4
lipophilicity	random	-0.326	-0.238	–	0.728	4	4
lipophilicity	scaffold	-0.219	-0.187	–	0.684	4	4

Table S14: Similarity-quartile sensitivity analysis. Quartile 1 is the lowest-similarity group and quartile 4 the highest-similarity group.

Endpoint	Split	Quartile	Model-seed cells	Mean similarity	Mean risk	Mean miscoverage	Unseen-scaffold rate
bbbp	cluster	1	40	0.273	0.230	0.157	0.758
bbbp	cluster	2	40	0.409	0.139	0.086	0.659
bbbp	cluster	3	40	0.529	0.122	0.065	0.504
bbbp	cluster	4	40	0.669	0.106	0.063	0.290
bbbp	random	1	80	0.291	0.201	0.161	0.736
bbbp	random	2	80	0.462	0.114	0.084	0.518
bbbp	random	3	80	0.621	0.102	0.077	0.407
bbbp	random	4	80	0.791	0.080	0.062	0.147
bbbp	scaffold	1	80	0.278	0.229	0.149	1.000
bbbp	scaffold	2	80	0.407	0.160	0.102	1.000
bbbp	scaffold	3	80	0.523	0.104	0.060	1.000
bbbp	scaffold	4	80	0.676	0.092	0.061	1.000
clintox	cluster	1	40	0.219	0.099	0.147	0.750
clintox	cluster	2	40	0.323	0.062	0.086	0.681
clintox	cluster	3	40	0.444	0.051	0.070	0.497
clintox	cluster	4	40	0.598	0.079	0.109	0.369
clintox	random	1	80	0.234	0.099	0.149	0.708
clintox	random	2	80	0.376	0.050	0.077	0.563
clintox	random	3	80	0.546	0.041	0.069	0.440
clintox	random	4	80	0.772	0.082	0.115	0.204
clintox	scaffold	1	80	0.228	0.123	0.154	1.000
clintox	scaffold	2	80	0.326	0.050	0.071	1.000
clintox	scaffold	3	80	0.441	0.060	0.078	1.000
clintox	scaffold	4	80	0.629	0.073	0.090	1.000
esol	cluster	1	20	0.290	1.373	0.143	0.425
esol	cluster	2	20	0.433	1.194	0.086	0.243
esol	cluster	3	20	0.520	1.062	0.061	0.200
esol	cluster	4	20	0.637	1.000	0.067	0.073
esol	random	1	40	0.299	1.342	0.161	0.438
esol	random	2	40	0.469	1.126	0.100	0.195
esol	random	3	40	0.583	0.944	0.054	0.102
esol	random	4	40	0.797	0.954	0.079	0.091
esol	scaffold	1	40	0.225	1.496	0.070	1.000
esol	scaffold	2	40	0.331	1.544	0.083	1.000
esol	scaffold	3	40	0.435	1.792	0.153	1.000
esol	scaffold	4	40	0.599	1.618	0.130	1.000
lipophilicity	cluster	1	20	0.305	1.369	0.187	0.870
lipophilicity	cluster	2	20	0.443	1.119	0.129	0.751
lipophilicity	cluster	3	20	0.572	0.948	0.078	0.653
lipophilicity	cluster	4	20	0.719	0.761	0.039	0.482
lipophilicity	random	1	40	0.330	1.273	0.211	0.830

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Table S14 (continued)

Endpoint	Split	Quartile	Model-seed cells	Mean similarity	Mean risk	Mean miscoverage	Unseen-scaffold rate
lipophilicity	random	2	40	0.551	0.962	0.115	0.582
lipophilicity	random	3	40	0.725	0.713	0.047	0.402
lipophilicity	random	4	40	0.857	0.566	0.032	0.200
lipophilicity	scaffold	1	40	0.308	1.284	0.171	1.000
lipophilicity	scaffold	2	40	0.457	1.096	0.127	1.000
lipophilicity	scaffold	3	40	0.618	0.913	0.073	1.000
lipophilicity	scaffold	4	40	0.770	0.727	0.029	1.000

Table S15: Threshold sensitivity for low-domain minus high-domain risk and miscoverage. Threshold partitions are sensitivity analyses, not universal applicability-domain definitions.

Endpoint	Split	Threshold	Valid cells	Low-domain fraction	Δ risk low-high	Δ miscov. low-high	Model CIs risk > 0	Model CIs miscov. > 0
bbbp	cluster	0.30	40	0.175	0.135	0.117	8	8
bbbp	cluster	0.40	40	0.354	0.082	0.067	8	5
bbbp	cluster	0.50	40	0.545	0.072	0.058	7	7
bbbp	cluster	0.60	40	0.801	0.059	0.038	6	2
bbbp	cluster	0.70	40	0.918	0.071	0.044	7	1
bbbp	cluster	0.80	40	0.985	0.094	0.076	4	8
bbbp	random	0.30	80	0.132	0.161	0.138	8	8
bbbp	random	0.40	80	0.282	0.095	0.078	8	8
bbbp	random	0.50	80	0.425	0.080	0.066	8	8
bbbp	random	0.60	80	0.585	0.062	0.047	8	8
bbbp	random	0.70	80	0.755	0.065	0.049	8	8
bbbp	random	0.80	80	0.913	0.052	0.040	8	7
bbbp	scaffold	0.30	80	0.166	0.158	0.109	8	8
bbbp	scaffold	0.40	80	0.358	0.107	0.070	8	8
bbbp	scaffold	0.50	80	0.569	0.093	0.063	8	8
bbbp	scaffold	0.60	80	0.770	0.073	0.043	8	7
bbbp	scaffold	0.70	80	0.932	0.078	0.037	8	5
bbbp	scaffold	0.80	80	0.985	0.106	0.062	6	5
clintox	cluster	0.30	40	0.330	0.040	0.057	7	5
clintox	cluster	0.40	40	0.535	0.014	0.025	0	0
clintox	cluster	0.50	40	0.733	-0.005	-0.003	0	0
clintox	cluster	0.60	40	0.908	-0.011	-0.006	0	0
clintox	cluster	0.70	40	0.970	0.039	0.057	0	1
clintox	cluster	0.80	40	0.993	0.007	0.025	0	0
clintox	random	0.30	80	0.249	0.044	0.066	8	7
clintox	random	0.40	80	0.417	0.019	0.033	3	5
clintox	random	0.50	80	0.559	0.003	0.008	0	2
clintox	random	0.60	80	0.696	-0.016	-0.014	0	0
clintox	random	0.70	80	0.819	-0.015	-0.019	0	0
clintox	random	0.80	80	0.932	-0.017	-0.027	0	0
clintox	scaffold	0.30	80	0.307	0.046	0.057	8	8
clintox	scaffold	0.40	80	0.538	0.019	0.029	2	6
clintox	scaffold	0.50	80	0.720	0.010	0.014	1	3
clintox	scaffold	0.60	80	0.865	-0.017	-0.014	0	0
clintox	scaffold	0.70	80	0.946	-0.039	-0.045	0	0
clintox	scaffold	0.80	80	0.987	-0.080	-0.076	0	0
esol	cluster	0.30	20	0.115	0.349	0.083	2	1
esol	cluster	0.40	20	0.300	0.262	0.061	3	3
esol	cluster	0.50	20	0.516	0.249	0.047	1	1
esol	cluster	0.60	20	0.862	0.167	0.013	0	1
esol	cluster	0.70	20	0.958	0.174	-0.005	1	1

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Table S15 (continued)

Endpoint	Split	Threshold	Valid cells	Low-domain fraction	Δ risk low-high	Δ miscov. low-high	Model CIs risk > 0	Model CIs miscov. > 0
esol	cluster	0.80	16	0.984	-0.128	-0.058	0	0
esol	random	0.30	40	0.108	0.483	0.114	4	4
esol	random	0.40	40	0.245	0.345	0.084	4	4
esol	random	0.50	40	0.411	0.306	0.074	4	4
esol	random	0.60	40	0.649	0.232	0.048	3	3
esol	random	0.70	40	0.817	0.183	0.025	3	1
esol	random	0.80	40	0.900	0.078	-0.003	1	1
esol	scaffold	0.30	40	0.295	-0.174	-0.051	0	0
esol	scaffold	0.40	40	0.545	-0.199	-0.067	0	0
esol	scaffold	0.50	40	0.753	0.013	-0.027	1	0
esol	scaffold	0.60	40	0.900	0.066	-0.048	1	0
esol	scaffold	0.70	40	0.968	0.297	-0.038	1	0
esol	scaffold	0.80	36	0.983	0.835	0.094	4	3
lipophilicity	cluster	0.30	20	0.108	0.499	0.124	4	3
lipophilicity	cluster	0.40	20	0.299	0.415	0.102	4	4
lipophilicity	cluster	0.50	20	0.469	0.396	0.102	4	4
lipophilicity	cluster	0.60	20	0.687	0.378	0.094	4	4
lipophilicity	cluster	0.70	20	0.865	0.403	0.091	4	4
lipophilicity	cluster	0.80	20	0.968	0.454	0.102	4	4
lipophilicity	random	0.30	40	0.081	0.600	0.167	4	4
lipophilicity	random	0.40	40	0.214	0.537	0.150	4	4
lipophilicity	random	0.50	40	0.319	0.532	0.149	4	4
lipophilicity	random	0.60	40	0.419	0.500	0.134	4	4
lipophilicity	random	0.70	40	0.564	0.466	0.117	4	4
lipophilicity	random	0.80	40	0.799	0.419	0.091	4	4
lipophilicity	scaffold	0.30	40	0.103	0.413	0.110	4	4
lipophilicity	scaffold	0.40	40	0.285	0.383	0.098	4	4
lipophilicity	scaffold	0.50	40	0.438	0.366	0.097	4	4
lipophilicity	scaffold	0.60	40	0.590	0.369	0.097	4	4
lipophilicity	scaffold	0.70	40	0.774	0.367	0.094	4	4
lipophilicity	scaffold	0.80	40	0.933	0.388	0.083	4	4

S12 Selective prediction

Table S16: Selective-prediction summaries against matched random rejection. Blank class-specific cells are not applicable to regression.

Endpoint	Task	Split	Rejection score	Primary pAURC impr.	Risk impr. at 0.50	Balanced pAURC impr.	Positive retention	Negative retention	Class- balance shift
bbbp	classification	cluster	1 - max predicted probability	0.070	0.090	0.060	0.562	0.312	0.091
bbbp	classification	cluster	1 - max Tanimoto to training set	0.029	0.035	0.075	0.479	0.570	-0.033
bbbp	classification	random	1 - max predicted probability	0.058	0.075	0.057	0.545	0.359	0.067
bbbp	classification	random	1 - max Tanimoto to training set	0.028	0.034	0.066	0.469	0.604	-0.049
bbbp	classification	scaffold	1 - max predicted probability	0.065	0.084	0.055	0.558	0.319	0.086
bbbp	classification	scaffold	1 - max Tanimoto to training set	0.036	0.048	0.071	0.497	0.519	-0.007
clintox	classification	cluster	1 - max predicted probability	0.021	0.028	-0.021	0.306	0.515	-0.028
clintox	classification	cluster	1 - max Tanimoto to training set	0.004	0.008	-0.004	0.440	0.504	-0.008
clintox	classification	random	1 - max predicted probability	0.026	0.034	-0.029	0.269	0.518	-0.030
clintox	classification	random	1 - max Tanimoto to training set	0.000	0.007	0.043	0.458	0.505	-0.006
clintox	classification	scaffold	1 - max predicted probability	0.028	0.036	-0.024	0.283	0.517	-0.032
clintox	classification	scaffold	1 - max Tanimoto to training set	0.007	0.009	0.012	0.461	0.503	-0.006
esol	regression	cluster	1 - max Tanimoto to training set	0.100	0.155	—	—	—	—
esol	regression	random	1 - max Tanimoto to training set	0.109	0.165	—	—	—	—
esol	regression	scaffold	1 - max Tanimoto to training set	-0.047	-0.101	—	—	—	—
lipophilicity	regression	cluster	1 - max Tanimoto to training set	0.213	0.258	—	—	—	—
lipophilicity	regression	random	1 - max Tanimoto to training set	0.250	0.321	—	—	—	—
lipophilicity	regression	scaffold	1 - max Tanimoto to training set	0.200	0.245	—	—	—	—

S13 Frozen-result integrity manifest

Table S17: Integrity manifest for the manuscript-ready frozen result tables. SHA-256 prefixes are printed for readability; full hashes are retained in the machine-readable integrity manifest.

File	Rows	Expected	Valid	SHA-256
results/manuscript_assets/tables/table_rq1_classification_calibration.csv	6	6	True	e59542f1a839...
results/manuscript_assets/tables/table_rq1_classification_performance.csv	6	6	True	8ed879bd820f...
results/manuscript_assets/tables/table_rq1_regression_performance.csv	6	6	True	548e39d3e775...
results/manuscript_assets/tables/table_rq2_ad_continuous.csv	12	12	True	3ae457990fc9...
results/manuscript_assets/tables/table_rq2_ad_similarity_quartiles.csv	48	48	True	920bdcc1bc65...
results/manuscript_assets/tables/table_rq2_ad_threshold_sensitivity.csv	72	72	True	59d2adf19d58...
results/manuscript_assets/tables/table_rq2_rq3_selective_prediction.csv	18	18	True	b3b11c56b03a...
results/manuscript_assets/tables/table_rq3_rq4_classification_conformal.csv	18	18	True	f71eb1f676a5...
results/manuscript_assets/tables/table_rq3_rq4_classification_paired_effects.csv	12	12	True	fba0a24bb7c2...
results/manuscript_assets/tables/table_rq4_regression_conformal.csv	18	18	True	1c80f2be1424...
results/manuscript_assets/tables/table_rq4_regression_paired_effects.csv	12	12	True	fe15246fe6ab...